

Implemented equations for $\Gamma^{\text{III}_c} / \chi^\theta$ AMC DIRECT, fact Path A (sixj), fact Path B (Pandya)

Code anchors in `imsrg`
`ReferenceImplementations.cc`, `FactorizedDoubleCommutator.eths.cc`

Abstract

Exact equations as implemented for the two-body piece Γ^{III_c} of $[\Omega, [\Omega, \Gamma]_{3b}]_{2b}$ in the ethS / TTS stack. Product ranks: Ω may be tensor (rank λ), Γ and Z are scalar; χ^θ is scalar ($T \times T \rightarrow S$). Ground truth = AMC/TTS DIRECT. Path A matches DIRECT term-by-term; Path B (Ile Pandya) does not.

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1 Notation

- Ω : anti-Hermitian generator (IMS RG η), J-rank λ . Reduced TBMEs via `GetTBME_J`.

- Γ : Hermitian scalar two-body operator.
- Z : Hermitian scalar output (additive piece of $\Gamma^{\text{III}c}$).
- Occupations: $n_a \in [0, 1]$, $\bar{n}_a = 1 - n_a$.
- P_{ij} : fermion exchange on legs (i, j) with the usual phase; code uses `ket Phase(J)` for $(1 - P)$.
- $\hat{J} = \sqrt{2J + 1}$, $\hat{\lambda} = \sqrt{2\lambda + 1}$.
- Equal- J channels only for scalar Z : δ_{J_0, J_1} .

Code map

Name	Routine	File
DIRECT / TTS	<code>comm223_232_tts.GIIIC(..., which_term)</code>	<code>ReferenceImplementations.cc</code>
Path A	<code>comm223_232_GIIIC</code> (default)	<code>FactorizedDoubleCommutator_eths.cc</code>
χ DGEMM	<code>FillChiThetaG3c_DGEMM</code>	same
χ loop	<code>FillChiThetaG3c</code>	same
Path B	<code>SetUse_TypeGIIIC_factorized(true)</code>	same

2 M-scheme (AMC input)

Occupation factor shared by both seeds:

$$f(a, b, c) = \bar{n}_a \bar{n}_b n_c + n_a n_b \bar{n}_c. \quad (1)$$

Full diagram:

$$\Gamma_{ijkl}^{\text{III}c} = -\frac{1}{2} \sum_{abcd} f(a, b, c) (1 - P_{ij})(1 - P_{kl}) \left(\underbrace{\Omega_{abcl} \Omega_{idab} \Gamma_{cjkd}}_{\text{Term1}} + \underbrace{\Omega_{icab} \Omega_{abdl} \Gamma_{djkc}}_{\text{Term2}} \right). \quad (2)$$

Split AMC inputs (also in `learn/amc_tts/input/dc_G3c_term{1,2}.txt`):

$$\Gamma_{T1}^{\text{III}c} = -\frac{1}{2} \sum_{abcd} f(a, b, c) P(i/j) P(k/l) \Omega_{abcl} \Omega_{idab} \Gamma_{cjkd}, \quad (3)$$

$$\Gamma_{T2}^{\text{III}c} = -\frac{1}{2} \sum_{abcd} f(a, b, c) P(i/j) P(k/l) \Omega_{icab} \Omega_{abdl} \Gamma_{djkc}. \quad (4)$$

3 AMC / DIRECT (J-coupled)

Source: `learn/amc_tts/output/G3c.tex`. Implemented in `comm223_232_tts.GIIIC` with `which_term` $\in \{0, 1, 2\}$ (both / Term1 / Term2). External legs written as $(p, g; q, h)$ with J_0 ; exchanges implement $(1 - P_{pg})(1 - P_{qh})$.

3.1 Term1 (maps to χ_k strip)

$$\begin{aligned}
Z_{pg,qh}^{(T1)J_0} &= -\frac{1}{2} \delta_{J_0J_1} \sum_{abcd J_2J_3J_4J_5} f(a, b, c) (-1)^{J_2+J_3+\lambda} \hat{J}_4^2 \hat{J}_5^2 \hat{\lambda}^{-1} \\
&\times \left\{ \begin{matrix} j_p & j_h & J_5 \\ j_c & j_d & J_3 \end{matrix} \right\} \left\{ \begin{matrix} j_q & j_g & J_5 \\ j_c & j_d & J_4 \end{matrix} \right\} \left\{ \begin{matrix} j_h & j_q & J_0 \\ j_g & j_p & J_5 \end{matrix} \right\} \\
&\times \Omega_{ab,ch}^{J_2J_3\lambda} \Omega_{pd,ab}^{J_3J_2\lambda} \Gamma_{cg,qd}^{J_4} + (pg \leftrightarrow \text{exch.}).
\end{aligned} \tag{5}$$

3.2 Term2 (maps to χ_j strip)

$$\begin{aligned}
Z_{pg,qh}^{(T2)J_0} &= -\frac{1}{2} \delta_{J_0J_1} \sum_{abcd J_2J_3J_4J_5} f(a, b, c) (-1)^{J_2+J_3+\lambda} \hat{J}_4^2 \hat{J}_5^2 \hat{\lambda}^{-1} \\
&\times \left\{ \begin{matrix} j_h & j_p & J_5 \\ j_c & j_d & J_2 \end{matrix} \right\} \left\{ \begin{matrix} j_g & j_q & J_5 \\ j_c & j_d & J_4 \end{matrix} \right\} \left\{ \begin{matrix} j_h & j_q & J_0 \\ j_p & j_g & J_5 \end{matrix} \right\} \\
&\times \Omega_{pc,ab}^{J_2J_3\lambda} \Omega_{ab,dh}^{J_3J_2\lambda} \Gamma_{dg,qc}^{J_4} + (pg \leftrightarrow \text{exch.}).
\end{aligned} \tag{6}$$

Note: Term1 keeps χ -like outer channel at J_3 for $(pd; ch)$; Term2's unfactored mid- J labels differ (J_2 on outer legs of the $\Omega\Omega$ product). Both retain independent J_4 on Γ and bridge J_5 .

4 Intermediate χ^θ (fact-slow / fact-fast)

Define the mid- J ladder product (AMC packaging; $\hat{J}$ absorbed downstream):

$$T_{ij,kl}^J = \sum_{J'} (-1)^{J+J'+\lambda} \hat{\lambda}^{-1} \sum_{ab} \Omega_{ij,ab}^{JJ'\lambda} \Omega_{ab,kl}^{J'J\lambda}. \tag{7}$$

With occupations on the *internal* pair (a, b) only through the weights $W_{pp}(ab) = \bar{n}_a \bar{n}_b$, $W_{hh}(ab) = n_a n_b$, one builds

$$T_{ij,kl}^{pp,J} = \sum_{J'} (-1)^{J+J'+\lambda} \hat{\lambda}^{-1} \sum_{ab} W_{pp}(ab) \Omega_{ij,ab}^{JJ'\lambda} \Omega_{ab,kl}^{J'J\lambda}, \tag{8}$$

$$T_{ij,kl}^{hh,J} = \sum_{J'} (-1)^{J+J'+\lambda} \hat{\lambda}^{-1} \sum_{ab} W_{hh}(ab) \Omega_{ij,ab}^{JJ'\lambda} \Omega_{ab,kl}^{J'J\lambda}. \tag{9}$$

DGEMM implements (8)–(9); the orbit loop `FillChiThetaG3c` writes the equivalent sum with f inside.

4.1 Two strips

$$\chi_k^J(ij, kl) = n_k T_{ij,kl}^{pp,J} + \bar{n}_k T_{ij,kl}^{hh,J} = \sum_{ab} f(a, b, k) [\Omega \otimes \Omega]_{ij,kl}^{(0)}, \tag{10}$$

$$\chi_j^J(ij, kl) = n_j T_{ij,kl}^{pp,J} + \bar{n}_j T_{ij,kl}^{hh,J} = \sum_{ab} f(a, b, j) [\Omega \otimes \Omega]_{ij,kl}^{(0)}. \tag{11}$$

So χ_k weights the **ket-first** index; χ_j weights the **bra-second** index.

4.2 Hermiticity (implemented / verified)

For real Ω (Hermitian or anti-Hermitian),

$$\chi_j = \chi_k^T, \quad \chi^\theta \equiv \chi_k + \chi_j = \chi_k + \chi_k^T \quad (\text{Hermitian}). \quad (12)$$

Each strip alone is neither Hermitian nor anti-Hermitian. Path B must use the sum; strips cannot be Pandya'd separately under the Factorized hermiticity assumptions.

5 Path A — fact-slow / production (matches DIRECT)

Same $(1 - P_{pg})(1 - P_{qh})$ exchanges as DIRECT. After χ absorbs (a, b) , J_2 , phase, and $\hat{\lambda}^{-1}$:

5.1 Term1

$$\begin{aligned} Z_{pg,qh}^{(A1)J_0} = & -\frac{1}{2} \sum_{cd J_3 J_4 J_5} \hat{J}_4^2 \hat{J}_5^2 \begin{Bmatrix} j_p & j_h & J_5 \\ j_c & j_d & J_3 \end{Bmatrix} \begin{Bmatrix} j_q & j_g & J_5 \\ j_c & j_d & J_4 \end{Bmatrix} \begin{Bmatrix} j_h & j_q & J_0 \\ j_g & j_p & J_5 \end{Bmatrix} \\ & \times \chi_k^{J_3}(p, d, c, h) \Gamma_{cg,qd}^{J_4} + \text{exchanges}. \end{aligned} \quad (13)$$

5.2 Term2

$$\begin{aligned} Z_{pg,qh}^{(A2)J_0} = & -\frac{1}{2} \sum_{cd J_3 J_4 J_5} \hat{J}_4^2 \hat{J}_5^2 \begin{Bmatrix} j_h & j_p & J_5 \\ j_c & j_d & J_3 \end{Bmatrix} \begin{Bmatrix} j_g & j_q & J_5 \\ j_c & j_d & J_4 \end{Bmatrix} \begin{Bmatrix} j_h & j_q & J_0 \\ j_p & j_g & J_5 \end{Bmatrix} \\ & \times \chi_j^{J_3}(p, c, d, h) \Gamma_{dg,qc}^{J_4} + \text{exchanges}. \end{aligned} \quad (14)$$

Verified: $Z^{(A1)} \equiv Z^{(T1)}$, $Z^{(A2)} \equiv Z^{(T2)}$, $Z^{(A1)} + Z^{(A2)} \equiv \text{DIRECT}$ (to $\sim 10^{-13}$).

6 Path B — AMC χ^θ + scalar Iie Pandya (experimental)

6.1 Pack Hermitian scalar χ

$$\chi_{ij,kl}^{\theta,J} = \chi_k^J(ij, kl) + \chi_j^J(ij, kl) \quad (\text{same slots}), \quad (15)$$

store as reduced-product ME, then `MakeNotReduced()` for scalar Factorized reuse (`ChiThetaToScalarOperator`).

6.2 Pandya (same formula as Factorized Iie)

$$\bar{X}_{ab,cd}^J = \sum_{J'} (-1)^{j_b+j_c+J'} \hat{J}'^2 \begin{Bmatrix} j_a & j_b & J \\ j_c & j_d & J' \end{Bmatrix} X_{ad,bc}^{J'}. \quad (16)$$

Applied to non-reduced χ^θ and $\Gamma \rightarrow \bar{\chi}, \bar{\Gamma}$.

6.3 CC product and inverse Pandya

$$\bar{\chi}_\Gamma = \bar{\chi} \bar{\Gamma}, \quad (17)$$

$$Z_{ij,kl}^{(B)J} = -\frac{1}{2} (1 - P_{ij})(1 - P_{kl}) \sum_{J'} \hat{J}'^2 \begin{Bmatrix} j_i & j_j & J \\ j_k & j_l & J' \end{Bmatrix} (\bar{\chi}_\Gamma)^{J'}_{il,kj}. \quad (18)$$

(Implementation follows Factorized Iie layout / 2n CC matrices; see `PackChiThetaFactLayout` and the Pandya block in `comm223.232.GIIIC` Path B.)

Verified: $Z^{(B)} \neq \text{DIRECT}$ even at $\lambda = 0$ (with Hermitian AMC $\chi_k + \chi_j$).

7 Contrast: scalar Factorized Iie (different oracle)

Factorized BruteForce Iie (scalar η), *not* AMC G3c:

$$\begin{aligned}
Z_{pg,qh}^{\text{Iie},J_0} = & -\frac{1}{2}P_{pg}P_{qh} \sum_{abcd,J_2,J_3,J_4} \hat{J}_2^2 \hat{J}_3^2 \hat{J}_4^2 (\bar{n}_b n_a n_c + \bar{n}_a \bar{n}_c n_b) \\
& \times \begin{Bmatrix} j_p & j_h & J_4 \\ j_b & j_d & J_2 \end{Bmatrix} \begin{Bmatrix} j_q & j_g & J_4 \\ j_b & j_d & J_3 \end{Bmatrix} \begin{Bmatrix} j_h & j_q & J_0 \\ j_g & j_p & J_4 \end{Bmatrix} \\
& \times \eta_{ac,bh}^{J_2} \eta_{pd,ac}^{J_2} \Gamma_{bg,qd}^{J_3}.
\end{aligned} \tag{19}$$

Packed χ uses *outer-leg* occupations:

$$\chi^{\text{IV}} = \eta \eta_{\text{occ},c} + (\eta \eta_{\text{occ},d})^T, \tag{20}$$

then (16)–(18). That package matches **Iie**, not AMC (5)–(6).

8 Why Path A $\neq$ Path B (same χ)

After factoring χ_k from Term1, Path A still contracts with

$$\sum_{cd,J_3,J_4,J_5} \hat{J}_4^2 \hat{J}_5^2 \left\{ \begin{array}{cc} \cdots & J_3 \\ J_5 & \chi \end{array} \right\} \left\{ \begin{array}{cc} \cdots & \\ J & 4 \end{array} \right\} \left\{ \begin{array}{c} /afterBmatrix \cdots \end{array} \right\} k^{J_3} \Gamma^{J_4}, \tag{21}$$

i.e. χ and Γ in **independent** J -channels bridged by J_5 .

Path B replaces the whole Term1+Term2 content by a single cross-coupled product (17)–(18) (the Iie identity). These are different angular-momentum identities. Hermiticity of $\chi_k + \chi_j$ is necessary for (16) as used in Factorized, but not sufficient to reproduce (21).

Object	Equation	Matches DIRECT?
DIRECT Term1/2	(5),(6)	(definition)
Path A	(13),(14)	yes
χ_k, χ_j	(10),(11)	(shared)
Path B	(15)–(18)	no
Factorized Iie	(19)–(20)	no (different diagram)

9 Debug switches

```

which_term: 0 = both, 1 = Term1 / chi_k, 2 = Term2 / chi_j
ReferenceImplementations.comm223_232_tts_GIIIC(Eta,Gamma,Z, which_term)
FactorizedDoubleCommutator_eths.SetUse_TypeGIIIC_which_term(t)
DebugCompareGIIICTerms(Eta, Gamma)    # Path A + Path B vs DIRECT

```